

MS4215: Lab 6

ANOVA and ANCOVA: Palmer Penguins

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Introduction

The dataset `penguins.csv` contains morphological measurements for 333 penguins collected at Palmer Station, Antarctica. The variables are:

Variable	Description
<code>species</code>	Species (Adelie, Chinstrap, Gentoo)
<code>island</code>	Island (Biscoe, Dream, Torgersen)
<code>bill_length_mm</code>	Bill length (mm)
<code>bill_depth_mm</code>	Bill depth (mm)
<code>flipper_length_mm</code>	Flipper length (mm) — outcome
<code>body_mass_g</code>	Body mass (g)
<code>sex</code>	Sex (female, male)

The lab has three parts. **Part A** examines whether flipper length differs by species using one-way ANOVA. **Part B** adds sex as a second factor using two-way ANOVA. **Part C** uses ANCOVA to test species differences after controlling for bill length.

Before attempting the questions, run the setup chunk below to load the required packages and data.

```
#install.packages("tidyverse")
#install.packages("emmeans")
library(tidyverse)
library(emmeans)

# In the line of code below, you will need to change the location to where you
# have saved the file
df <- read_csv("data/penguins.csv")
df$species <- factor(df$species)
df$sex <- factor(df$sex)
```

Questions

Part A: One-way ANOVA

(i) Explore the data

Use `summary()` and `count()` to describe the dataset. Create a boxplot of `flipper_length_mm` by species using `ggplot2`. Compute the mean and SD of flipper length for each species using `group_by()` and `summarise()`. Comment on any apparent differences between species.

(ii) One-way ANOVA and post-hoc comparisons

Fit a one-way ANOVA:

```
m1 <- aov(flipper_length_mm ~ species, data = df)
```

Report the F statistic, degrees of freedom, and p-value from `summary(m1)`. Then apply post-hoc comparisons using:

- `TukeyHSD(m1)` for all pairwise Tukey comparisons
- `pairs(emmeans(m1, ~ species), adjust = "tukey")` for the same test via the `emmeans` interface

Which pairs of species differ significantly? What are the estimated differences?

(iii) Estimated marginal means

Use `emmeans(m1, ~ species)` to extract estimated marginal means and 95% confidence intervals for each species. Plot the means and error bars using `ggplot2`. Describe in plain language how the three species differ in flipper length.

Part B: Two-way ANOVA

(iv) Two-way ANOVA

Fit a two-way ANOVA including both species and sex and their interaction:

```
m2 <- aov(flipper_length_mm ~ species * sex, data = df)
```

Report all three F-tests from `summary(m2)`. Is the interaction term significant? What does this mean?

(v) Interaction plot and simple effects

Use `emmip(m2, sex ~ species, CIs = TRUE)` to produce an interaction profile plot. Are the lines parallel? Use `joint_tests(m2, by = "sex")` to test whether species differences are significant within each sex separately.

Part C: ANCOVA

(vi) Adjusting for bill length

Mean bill length differs substantially across species. Fit an ANCOVA that controls for `bill_length_mm` as a continuous covariate while testing for species differences in flipper length:

```
m_ancova <- lm(flipper_length_mm ~ bill_length_mm + species, data = df)
m_interact <- lm(flipper_length_mm ~ bill_length_mm * species, data = df)
```

- Use `anova(m_ancova, m_interact)` to test the parallel slopes assumption
- Report the coefficient and p-value for `bill_length_mm`
- Extract adjusted species means from `emmeans(m_ancova, ~ species)`

- Compare these adjusted means with the unadjusted means from Part A(i). Which direction did the species estimates shift, and why?